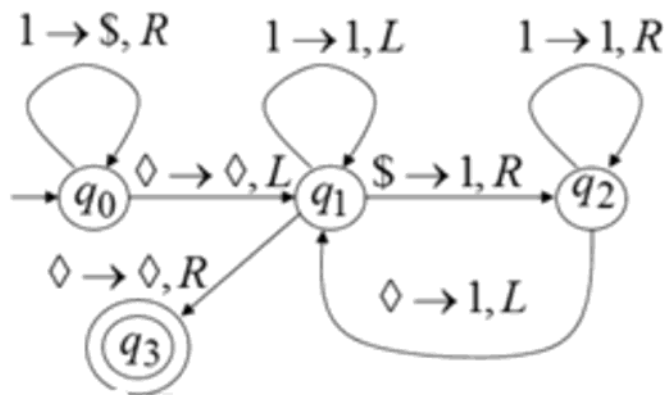



 $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$
 $Q = \{q_0, q_1, q_2, q_3, q_4\}$
 $\Sigma = \{1\}$
 $\Gamma = \{1, \$, \diamond\}$
 $q_0 = q_0$
 $B = \diamond$
 $F = \{q_3\}$

Fungsi transisi:

 $\delta(q_0, 1) = (q_0, \$, R)$
 $\delta(q_0, \diamond) = (q_1, \diamond, L)$
 $\delta(q_1, \$) = (q_2, 1, R)$
 $\delta(q_1, 1) = (q_1, 1, L)$
 $\delta(q_1, \diamond) = (q_3, \diamond, R)$
 $\delta(q_2, 1) = (q_2, 1, R)$
 $\delta(q_2, \diamond) = (q_1, 1, L)$

- a. Tunjukkan beberapa string yang diterima oleh Mesin Turing tersebut dalam bentuk Deskripsi Instan!
- b. Tunjukkan ada fungsi komputasi yang diterima oleh Mesin Turing tersebut !

a)

Accept Input

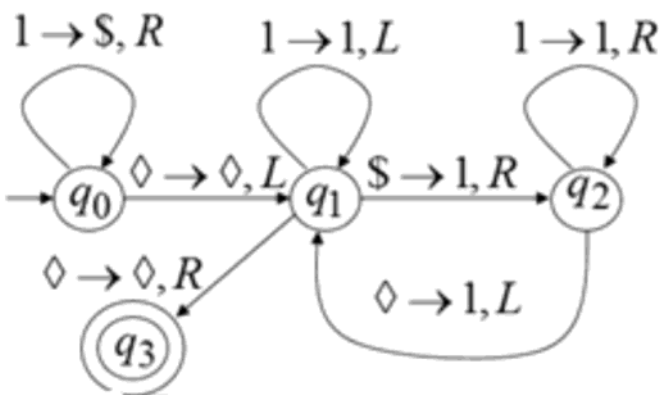


If machine halts
in a final state

Reject Input



If machine halts
in a non-final state
or
If machine enters
an infinite loop



String yang diterima: 1

$\diamond$	$\diamond$	1	$\diamond$	$\diamond$	$\diamond$	$\diamond$
------------	------------	---	------------	------------	------------	------------

 $\uparrow q_0$
 $q_0 1 >$

$\diamond$	$\diamond$	\$	$\diamond$	$\diamond$	$\diamond$	$\diamond$
------------	------------	----	------------	------------	------------	------------

 $\uparrow q_0$
 $\$ q_0 >$

$\diamond$	$\diamond$	\$	$\diamond$	$\diamond$	$\diamond$	$\diamond$
------------	------------	----	------------	------------	------------	------------

 $\uparrow q_1$
 $q_1 \$ >$

$\diamond$	$\diamond$	1	$\diamond$	$\diamond$	$\diamond$	$\diamond$
------------	------------	---	------------	------------	------------	------------

 $\uparrow q_2$
 $1 q_2 >$

$\diamond$	$\diamond$	1	1	$\diamond$	$\diamond$	$\diamond$
------------	------------	---	---	------------	------------	------------

 $\uparrow q_1$
 $1 q_1 1 >$

$\diamond$	$\diamond$	1	1	$\diamond$	$\diamond$	$\diamond$
------------	------------	---	---	------------	------------	------------

 $\uparrow$
 $q_1 \$ 11 >$

$\diamond$	$\diamond$	1	1	$\diamond$	$\diamond$	$\diamond$
------------	------------	---	---	------------	------------	------------

 $\uparrow$
 $q_3 11$
 $q_0 w > \dots$
 $q_0 11 >^* q_3 1111$
 $q_0 111 >^* q_3 11111$
b) $f(x) = 2x$
 $1 \rightarrow f(1) = 2$

$$I \rightarrow f(1) = 2$$

$$II \rightarrow f(2) = 4$$

$$III \rightarrow f(3) = 6$$